

Best Ollama embedding models for Spanish technical RAG

BGE-M3 is the strongest untested candidate, with the highest multilingual retrieval scores on MIRACL (69.2 nDCG@10), [\(nomic\)](#) a true encoder architecture, 8K context, and 3.2 million Ollama downloads proving stability. But this research uncovered a sobering reality: only **4 models in the entire Ollama ecosystem** meet all five hard requirements for Spanish technical document retrieval. The embedding model landscape on Ollama is far more constrained than expected — most multilingual models cap at 512 tokens, and most long-context models are English-only. A purpose-built Spanish model from Jina and Google's newest sub-500M champion round out the viable options, while a smaller Qwen3 variant offers a speed-optimized alternative from a family already proven on this dataset.

The 4 candidates ranked by likelihood of beating 0.770

The ranking below prioritizes models with strong evidence of Spanish retrieval capability, true encoder architectures, and practical Ollama deployment characteristics. One critical finding from the benchmark analysis: **BGE-M3 scores worst among open-source models on out-of-domain CLEF benchmarks** [\(arXiv\)](#) (41.0) despite leading MIRACL (69.2). Since your fiscal/tax documents are far from Wikipedia, CLEF-style out-of-domain performance may matter more than MIRACL scores — exactly as your Snowflake Arctic experience demonstrated.

#1 — BGE-M3: the multilingual retrieval heavyweight

Attribute	Value
Model name	bge-m3
Dimensions	1,024
Context window	8,192 tokens
Parameters	567M
Architecture	XLM-RoBERTa (BERT encoder, non-causal)
MIRACL avg (18 langs)	69.2 (dense) / 70.0 (hybrid)
MIRACL-O (EN/FR/ES/DE)	56.8 nDCG@10
CLEF out-of-domain	41.0 nDCG@10 (lowest among peers — potential red flag)
Ollama command	ollama pull bge-m3
Size on disk	1.2 GB (F16)
Known issues	None — battle-tested with 3.2M downloads

BGE-M3 from BAAI (Ollama) is the most downloaded multilingual embedding model on Ollama (Ollama) and holds the **highest MIRACL multilingual retrieval score** among all open-source models its size. It supports 100+ languages with Spanish explicitly included in training and evaluation. The XLM-RoBERTa encoder backbone with 24 transformer layers and CLS pooling (Ollama +2) is the gold standard for multilingual embeddings.

The model uniquely supports three retrieval modes — dense, sparse (learned lexical weights), and multi-vector (ColBERT) (Hugging Face +3) — though **only dense retrieval works through Ollama's API**. For hybrid retrieval, you'd need the Python FlagEmbedding library. (Deep Infra) Its 1,024 dimensions (Ollama) sit in your ideal sweet spot, and 8K context gives 16x headroom over your largest chunks. (Ollama)

The honest risk: BGE-M3 scores **41.0 on CLEF** (out-of-domain European language retrieval on news articles), the worst among comparable models. (Snowflake) This suggests possible overfitting to MIRACL's Wikipedia training domain. Your ARCA/AFIP technical documentation is about as far from Wikipedia as possible, so this model may underperform its benchmark promise — though likely not as catastrophically as Snowflake Arctic did. Community quantized versions are available via (ollama pull qllama/bge-m3:q8_0) (635 MB) or (qllama/bge-m3:q4_k_m) (438 MB) (Ollama) for faster initial embedding of all 2,304 chunks.

Expected performance: 0.77–0.82 on your dataset. The strong MIRACL-O score for European languages (56.8) and proven XLM-RoBERTa backbone make this the most probable model to beat your 0.770 baseline.

#2 — Jina Embeddings v2 Base ES: purpose-built for Spanish

Attribute	Value
Model name	(jina/jina-embeddings-v2-base-es)
Dimensions	768
Context window	8,192 tokens
Parameters	161M
Architecture	JinaBERT (BERT encoder with ALiBi positional encoding)
MIRACL score	Not independently benchmarked on MIRACL-18
Spanish retrieval	Jina's MTEB-es benchmark claims it outperforms BGE-M3 on Spanish retrieval
Ollama command	(ollama pull jina/jina-embeddings-v2-base-es)
Size on disk	~323 MB (F16)
Known issues	None — official Jina namespace, 17.5K downloads

This is the **only embedding model on Ollama specifically trained for Spanish**. Released by Jina AI in February 2024, it was built as a bilingual Spanish-English model trained to handle mixed-language content without bias toward either language. [Ollama](#) Jina's own benchmarks (via their MTEB-es evaluation suite) claim it outperforms both BGE-M3 and multilingual-e5-base on Spanish retrieval and clustering tasks. [jina](#)

At just **161M parameters and ~323 MB**, it would embed your 2,304 chunks roughly **3.5x faster** than BGE-M3. The 768-dimensional output matches your nomic-embed-text baseline exactly, making it a drop-in replacement in Qdrant without reconfiguring your collection schema. ALiBi positional encoding enables the 8K context window [Ollama](#) [Ollama](#) without the memory overhead of full position embeddings.

The honest risk: This is a smaller, older model (Feb 2024) with bilingual-only coverage. Jina's benchmark claims are self-published and not independently verified. The v2 architecture is now superseded by v3, and ALiBi is considered inferior to RoPE for long sequences. [Jina AI](#) However, for your 100% Spanish use case, a Spanish-optimized smaller model may outperform a larger generic multilingual model — this is the core bet here.

Expected performance: 0.76–0.81 on your dataset. The Spanish-specific training could give it an edge on ARCA terminology, SOAP specs, and fiscal vocabulary that generic multilingual models miss. The biggest unknown is whether 161M parameters provide enough capacity for technical domain vocabulary.

#3 — EmbeddingGemma: Google's efficient multilingual newcomer

Attribute	Value
Model name	embeddinggemma
Dimensions	768 (MRL truncatable to 512/256/128)
Context window	2,048 tokens (exactly meets threshold)
Parameters	308M
Architecture	Gemma 3 with bidirectional attention (encoder-style)
MMTEB ranking	#1 text-only multilingual model under 500M at release
Ollama command	ollama pull embeddinggemma
Size on disk	622 MB (BF16); 239 MB (QAT q4_0)
Known issues	Requires Ollama v0.11.10+ ; 2K context is borderline

Google's newest embedding model [Hugging Face](#) [ollama](#) (released late 2025) claimed the **top spot on MMTEB's multilingual leaderboard for models under 500M parameters**, outperforming even Qwen3-Embedding-0.6B on several multilingual task types. [arXiv](#) Built on the Gemma 3 architecture with

bidirectional attention explicitly enabled (`causal: false`), `Ollama` it functions as a true encoder despite its decoder-family origins. It supports 100+ languages and was trained on **320 billion tokens**. `Hugging Face +2`

The **2,048-token context window is the key concern**. `Ollama` Your chunks range from 300–500 tokens, so individual chunks fit comfortably. But there is zero headroom — any chunk that runs slightly over 2,048 tokens gets truncated silently. Matryoshka Representation Learning support lets you truncate dimensions to 512 or 256 `Hugging Face` `huggingface` if you want faster cosine similarity computation in Qdrant. Quantized variants drop the size to just **239 MB** (`q4_0`), `Ollama` making this by far the smallest deployment option.

Expected performance: 0.75–0.79 on your dataset. The MMTEB multilingual leadership is encouraging, but the model is very new (limited community validation), and the tight context window introduces a truncation risk that could silently degrade retrieval quality on your longer chunks.

#4 — Qwen3 Embedding 0.6B: the untested sibling of your best scorer

Attribute	Value
Model name	<code>qwen3-embedding:0.6b</code>
Dimensions	1,024 (MRL: 32–4,096)
Context window	32,768 tokens
Parameters	~600M
Architecture	Qwen3 decoder adapted for embedding (last-token pooling)
MMTEB multilingual	64.33 (Mean Task); 64.64 (Retrieval)
Ollama command	<code>ollama pull qwen3-embedding:0.6b</code>
Size on disk	639 MB (<code>Q8_0</code>)
Known issues	Decoder-based (uses causal attention, not bidirectional)

You already proved the Qwen3-Embedding family works well on your data — the 4B variant scored **0.802** (your best result). The 0.6B variant is the same architecture scaled down to roughly one-seventh the parameters. `Ollama` On MMTEB benchmarks, the 0.6B scores ~7% **lower on retrieval** than the 4B (64.64 vs 69.60), which projects to approximately **0.74–0.76** on your dataset — right around your current baseline.

This candidate makes strategic sense not as a performance upgrade but as a **speed-focused alternative**. At 639 MB `ollama` versus the 4B's 2.5 GB, `Ollama` initial embedding of 2,304 chunks will be significantly faster. The 32K context window `ollama` is massive overkill but eliminates any truncation concern. Matryoshka support lets you set dimensions to 768 or 1,024 `Ollama` to match your Qdrant collection `Ollama` without the 2,560-dimension overhead of the 4B variant.

Architecture caveat: This is a decoder-only model using causal (unidirectional) attention and last-token pooling, not a true bidirectional encoder. Despite this, it achieves higher MMTEB multilingual scores (64.33) than BGE-M3 (59.56) on the overall leaderboard, so the architectural disadvantage is offset by scale and training quality.

Expected performance: 0.74–0.76, likely just matching your nomic baseline rather than beating it. The value proposition is speed parity with better multilingual grounding, not a quality leap.

Why only 4 models qualify — and none may be a slam dunk

The Ollama embedding ecosystem has a severe gap at the intersection of **long context** + **multilingual** + **stable GGUF**. Here is why dozens of seemingly viable models fail your requirements:

- **512-token ceiling kills most multilingual encoders:** paraphrase-multilingual (278M), [Ollama](#) granite-embedding (278M), [Ollama](#) multilingual-e5-large (559M), nomic-embed-v2-moe (475M), [Nomic AI](#) and mxbai-embed-large all cap at 512 tokens. These are otherwise excellent models trapped by their BERT-era positional embedding limits.
- **GTE-Qwen2 has broken bidirectional attention in GGUF:** The GTE-Qwen2-1.5B model [Ollama](#) on Ollama (community upload) uses a Qwen2 decoder [Ollama](#) with a [causal=False](#) hack for bidirectional attention. In llama.cpp's GGUF runtime, causal attention is **hardcoded to True** for Qwen2, meaning the model silently degrades to unidirectional attention [GitHub](#) — making it worse than purpose-built decoders like Qwen3-Embedding.
- **Jina v3 isn't really on Ollama:** The [lazarevill/jina-embeddings-v3](#) community page shows "no models pushed." [Ollama](#) Even if it were uploaded, GGUF conversion strips the task-specific LoRA adapters that define v3's architecture. [arXiv](#) [GitHub](#)
- **GTE-multilingual-base (the ideal candidate) has no GGUF:** Alibaba's true-encoder GTE with 8K context and 70+ languages [Alibaba Cloud Community](#) [Hugging Face](#) simply hasn't been converted to GGUF format. Its custom mGTE architecture with RoPE and GLU complicates automated conversion. [Alibaba Cloud Community](#)

Your Snowflake Arctic experience (0.640 despite 66.8 MTEB) is not an outlier — it is the expected outcome when applying English-optimized or Wikipedia-domain models to Argentine fiscal technical documentation. The CLEF out-of-domain benchmark confirmed this pattern: models that score highest on MIRACL (Wikipedia-based) often score lowest on CLEF (news-based). [Snowflake](#) Your ARCA/AFIP SOAP documentation is a third domain entirely.

Recommended testing order and strategy

Test these models in the order below, stopping when you find one that meaningfully beats 0.770. Each subsequent test has diminishing expected returns.

Priority 1 — `ollama pull bge-m3` (1.2 GB). Highest probability of improvement. Run your full benchmark suite first. If it scores ≥ 0.79 , it is likely your best option. If it scores below 0.77, the CLEF overfitting concern is validated and you should test #2.

Priority 2 — `ollama pull jina/jina-embeddings-v2-base-es` (~323 MB). The wildcard. If BGE-M3 disappoints on Spanish technical queries, this Spanish-specialist model may capture domain vocabulary better despite being smaller. Its tiny size means you can embed all 2,304 chunks in minutes.

Priority 3 — `ollama pull embeddinggemma` (622 MB, requires Ollama $\geq$ v0.11.10). Worth testing if neither BGE-M3 nor Jina v2-es beats baseline. The newest model with the freshest training data. Verify your longest chunks don't exceed 2,048 tokens before committing.

Priority 4 — `ollama pull qwen3-embedding:0.6b` (639 MB). Only test this if you want a faster embedding pipeline at roughly baseline quality. Set `num_dims: 1024` via the Ollama API to avoid the default 4,096-dimension overhead.

If none of these beat your qwen3-embedding:4b score of 0.802, that model remains your best option — the 4.1% improvement with its mixed procedural query behavior may be an acceptable trade-off given the alternatives.

Conclusion

The honest takeaway is that **BGE-M3 is your highest-probability upgrade**, but expectations should be tempered. Your 0.770 baseline with nomic-embed-text v1 (an English-focused model) on 100% Spanish technical documents is already surprisingly strong, which suggests your chunking strategy and query formulation are doing heavy lifting. The gap between benchmark-predicted and actual performance on niche Spanish technical content can be **20+ percentage points** (as Snowflake Arctic proved), so no model can be recommended with high confidence without empirical testing on your specific dataset. The Jina v2 Spanish model (`Ollama`) at 161M parameters is the most intriguing dark horse — a purpose-built bilingual model may outperform multilingual generalists on ARCA fiscal documentation in ways that no benchmark captures.